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AMENDMENT NO. 2 AUGUST 2009
TO

**IS 4989 (PART 4) : 2003 MULTIPURPOSE AQUEOUS
FILM FORMING FOAM LIQUID CONCENTRATE FOR EXTINGUISHING
HYDROCARBON AND POLAR SOLVENT FIRES — SPECIFICATION**

(Page 1, clause 3) — Substitute the following for the existing:

- a) Type 3 × 6:
 - i) For Polar solvents at the rate of 6 percent concentration, and
 - ii) For hydrocarbons at the rate of 3 percent concentration.
- b) Type 3 × 3:
 - i) For Polar solvents at the rate of 3 percent concentration, and
 - ii) For hydrocarbons at the rate of 3 percent concentration.

[Page 2, Table 2 (see also Amendment No. 1)]— Substitute the following for the existing table:

Table 2 Performance Requirements

Sl No.	Performance	Requirements		Ref to Test in Annex
		Polar Solvent	<i>n</i> -heptane	
(1)	(2)	(3)	(4)	(5)
i)	Expansion at 27 ± 5°C	6 Min	6 Min	F
ii)	25 percent drainage time at 27 ± 5°C	5 min, Min 6%	3 min, Min 3%	G
iii)	Fire control time	120 s, Max	60 s, Max	H and J
iv)	Fire extinction time	300 s, Max	90 s, Max	H and J
v)	Burn Back	8 min, Min	10 min, Min	K

(Page 2, clause 8) — Add the following new clause and renumber the existing clause 8 as clause 9:

‘8 SAMPLING OF THE FOAM COMPOUND CONCENTRATE

8.1 The number of filled containers to be sampled from each batch for ascertaining conformity shall be in accordance with the following table:

<i>Scale of Sampling</i>	
Total Number of Filled Containers from Each Batch	No. of Containers to be Selected
Up to 100	5
101 - 200	10
201 and above	15

8.2 The containers shall be selected at random from each batch. To ensure the randomness of selection, method given in IS 4905 may be adopted. The equal amount of samples collected from each selected container should be mixed to make a composite sample to carry out the single conformity test.’

[Page 2, clause 8.1(b)] — Substitute the following for the existing:

- ‘b) Multipurpose foam for fire fighting AR-AFFF Type 3 × 6 or AR-AFFF Type 3 × 3 (as applicable),’

(Page 2, Annex B, clause B-1.2) — Add the following matter at the end of the clause:

‘or specific gravity bottle method’.

(Page 5, Annex D, clause D-5.1.1) — Substitute ‘ γ_b = surface tension of premixed foam solution as determined’ for ‘ γ_b = surface tension of premixed AFFF solution as determined’.

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(Page 5, Annex D, clause **D-5.1.1**) — Substitute ' γ_1 = inter-facial tension between premixed foam solution and cyclohexane as determined' for ' γ_1 = inter-facial tension between premixed AFFF and cyclohexane as determined'.

(Page 5, Annex E, clause **E-1**) — Substitute '11 l/min or 7.5 l/min at 11 kgf/cm² or 7.5 kgf/cm²' for '7.5 l per minute at 7 kgf/cm²'.

(Page 5, Annex E, clause **E-1**, last sentence) — Add the following at the end of clause:
'and Fig. 3.'

(Page 6, Annex E, Fig. 2) — Insert the new Fig. 3 given at page 3 of this Amendment after Fig. 2 and renumber all subsequent figures and their corresponding references in the text.

(Pages 6 and 7, Fig. 2, Captions) — Substitute 'FOAM MAKING NOZZLE of CAPACITY 7.5 l/min' for 'FOAM MAKING NOZZLE of CAPACITY 7.0 l/min'.

(Page 10, Annex H, clause **H-0**, Note 2, line 5) — Substitute 'comply requirements' for 'not differ'.

(Page 10, Annex H, clause **H-1.1**) — Substitute 'Foam solution temperature $27 \pm 5^\circ\text{C}$ ' for 'Foam solution temperature $27 \pm 1^\circ\text{C}$ '.

(Page 10, Annex H, clause **H-2**) — Substitute the following for the existing first sentence:

'Prepare a foam solution, mixing 94/97 l of potable or sea water as per requirement and 6/3 l of foam concentrate depending upon the type respectively. Fill up the solution in the foam tank of 100 l capacity and arrange for fire test.'

(Page 10, Annex H, clause **H-4**) — Delete 'acetone.....purchaser'.

(Page 10, Annex H, clause **H-7**) — Substitute 'Procedure given in Annex K' for 'Same as in IS 4989 (Part 1)'.

(Page 11, Annex J, clause **J-2**, line 1) — Substitute 'Pour 225 l of potable water and 50 l of *n*-Heptane' for 'Pour 100 l of isopropyl alcohol in the tank'.

(Page 11, Annex J, clause **J-2**, line 4) — Substitute '10 cm' for '30 cm' and add the following note at the end:

'NOTE — Water is not used for fire test of polar solvent foams.'

[Page 11, Annex J, clause **J-3** (see also Amendment No. 1)] — Substitute 'iso propyl alcohol' for '*n*-heptane' and '6 percent' for '3 percent'.

(Page 12, Annex K, clause **K-3**, line 1) — Substitute '*n*-heptane' for '73 NL petrol'.

(Page 12, Annex K, clause **K-3**, line 2) — Add the following at the end of first sentence:
'using 3 percent premixed solution'

All dimensions in millimeters.

FIG. 3 FOAM MAKING NOZZLE OF CAPACITY 1 l/min

(CED 22)

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